

Generation Interconnection System Impact Study Report

For Queue Project Q22-2139

Rose Garden 138 kV

August 31, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2139. The Interconnection Customer proposes a 113.0 MW solar photovoltaic facility interconnecting at the Rose Garden 138 kV substation in the Pennsylvania area. The request was received on July 21, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 441 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 12 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$13,140,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Fap / Westraver 138.0 kV	Kempster / Wfec Genco Llc 138.0 kV L	117.6	179	6.7
Kempster / Pueblo 138.0 kV Line	Tuco Substation / John Day 138.0 kV	117.2	235	5.3
Coffeyville / Belridge 1B 138.0 kV L	Wfec Genco Llc / Kempster 138.0 kV	116.7	195	6.7
Belridge 1B / Pueblo 138.0 kV	John Day / Tuco 138.0 kV	116.2	227	5.3
Coffeyville Substation / Fap 138.0 kV	Wfec Genco Llc / Kempster Substation 138.0 kV	110.9	194	6.7
Shaw / Wineagle 138.0 kV	Coffeyville / Belridge 1B 138.0 kV	105.3	274	7.6
Wineagle / Augusta Electric Plant No 1 138.0 kV	Medora / P.T.S 345.0 kV	104.8	349	9.9
Shaw / Wfec Genco Llc 138.0 kV	Belridge 1B / Coffeyville 138.0 kV	104.7	291	7.6
Wfec Genco Llc Substation / Kempster 138.0 kV	Belridge 1B / Coffeyville 138.0 kV	104.6	295	7.6
Westraver / Tenaska Westmoreland Generating Sta 138.0 kV	Wfec Genco Llc / Kempster 138.0 kV line	102.7	197	6.7
Mosaic Co Bartow Facility / Rose Garden 138.0 kV	Tuco / Medora 138.0 kV	101.6	175	32.5
North Allegheny Windpower Project / Wahiawa 345.0 kV	Sunrise Center / Merret 500.0 kV	100.4	1,717	5.4

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Westraver / Fap 138.0 kV	Rebuild line with higher-capacity conductor	\$4,378,000	42% (\$1,860,000)
Kempster / Pueblo 138.0 kV Line	Rebuild line with higher-capacity conductor	\$1,986,000	100% (\$1,986,000)
Coffeyville / Belridge 1B 138.0 kV	Rebuild line with higher-capacity conductor	\$1,986,000	42% (\$844,000)
Belridge 1B / Pueblo 138.0 kV	Rebuild line with higher-capacity conductor	\$2,392,000	100% (\$2,392,000)
Coffeyville Substation / Fap 138.0 kV	Rebuild line with higher-capacity conductor	\$3,554,000	42% (\$1,510,000)
Shaw / Wineagle 138.0 kV	Reconductor line	\$2,654,000	43% (\$1,138,000)
Wineagle / Augusta Electric Plant No 1 138.0 kV Line	Reconductor line	\$1,065,000	36% (\$386,000)
Shaw / Wfec Genco Llc 138.0 kV	Reconductor line	\$661,000	43% (\$283,000)
Wfec Genco Llc / Kempster Substation 138.0 kV Line	Reconductor line	\$1,770,000	43% (\$759,000)
Tenaska Westmoreland Generating Station / Westraver 138.0 kV line	Reconductor line	\$1,107,000	42% (\$470,000)
Mosaic Co Bartow Facility / Ros Garden 138.0 kV	Reconductor line	\$1,809,000	66% (\$1,201,000)
North Allegheny Windpower Project / Wahiawa 345.0 kV	Reconductor line	\$10,462,000	3% (\$311,000)
	Total allocated to project		\$13,140,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.